

Madhusudan Bhukta

Backend & AI-Native Software Developer

madhusudanbhukta.dev@gmail.com — github.com/megabytis — linkedin.com/in/madhusudan-bhukta — Portfolio

SUMMARY

Backend Developer specializing in scalable REST APIs, modular architectures, authentication systems, and high-performance backend engineering. Experienced with Node.js, Express, MongoDB, Redis, and integrating AI workflows. Improved API latency, optimized data access patterns, and implemented production-style features such as caching, RBAC, payment workflows, and Retrieval-Augmented Generation (RAG) pipelines.

SKILLS

Languages: JavaScript (ES6+), Python, SQL, TypeScript (learning)

Backend: Node.js, Express.js, REST API Architecture

Databases: MongoDB, Mongoose, Redis, MySQL (learning), ChromaDB (Vector)

Auth & Security: JWT, Refresh Tokens, RBAC

Engineering & AI: Caching, Rate Limiting, Pagination, Indexing, Domain Modeling, RAG Pipelines, Hybrid Search

Tools: Git, GitHub, Postman, Render, Vercel, Docker, Load Testing (Autocannon/k6), Linux (Arch)

PROJECTS

RAG-DocumentQnA (AI-Powered Document Search Engine) *Node.js, Python, ChromaDB, AI Agents*
github.com/megabytis/rag-documentqna

- Engineered a dual-service architecture integrating a Node.js API gateway with a Python AI orchestration backend.
- Built an advanced RAG pipeline supporting PDF, TXT, and DOCX document parsing and chunking.
- Reduced LLM hallucinations by optimizing context retrieval using Hybrid Search (Dense + BM25) and a Cross-Encoder Re-ranker.
- Designed scalable backend pipeline for handling multiple concurrent document queries efficiently.

ShopNexus (E-Commerce Backend Platform) *Node.js, Express, MongoDB, Redis, Stripe*
github.com/megabytis/shopnexus

- Improved product and category API latency by 35–60% using Redis caching and reduced redundant DB reads.
- Implemented JWT authentication, refresh tokens, RBAC, modular services, and full Stripe payment lifecycle.
- Designed domain models, implemented cache invalidation strategy, and performed load testing for system stability.

LearnSphere (Learning Management System) *React, Node.js, Express, MongoDB, Stripe*
github.com/megabytis/LearnSphere

- Built a full-stack LMS with role-based access control for Students, Instructors, and Admins.
- Implemented secure JWT-based authentication with cookie handling and session persistence.
- Integrated Stripe Checkout for course enrollment payments and dynamic order management.

DevTinder (Developer Matchmaking Platform) *Node.js, Express, MongoDB*
github.com/megabytis

- Built user authentication, profile system, and developer connection/matching logic.
- Structured application with modular backend architecture for scalability.

EXPERIENCE

Independent Backend Developer

2024–Present

- Built production-style backend systems with authentication, caching, and payment workflows.
- Improved API performance through caching, indexing, and optimized query patterns.
- Created middleware for validation, logging, error handling, and security.
- Performed debugging, load testing, and architecture refinement across multiple projects.

EDUCATION

Bachelor of Computer Applications (BCA), 2024–2027

NIIS Institute of Business Administration, Bhubaneswar, Odisha, India